LAZA Bronze Raw Ingestion

Version	00
Date	08/06/2026
Target audience	Data Architect, Data Engineer, Analytics Engineer, Data Analyst, QA Engineer and Product Owner
Objective	Consolidate the Bronze Raw Ingestion standard with the catalog and detailed specifications of the Validation indicators

A. Bronze Raw Ingestion functional standard

1. Overview and Context

The Bronze layer is the technical landing zone. It must store original files, versions, rows, cells, PDF pages, extracted data and raw observations. Bronze does not semantically correct the data; it preserves what was received, with sufficient traceability for audit and reprocessing.

Summary	Implement ingestion raw for files XLSX/XLS/PDF/API/web scraping and record metadata mandatory from the source, batch, file and extraction.
Objective	Garantir that no data between in the LAZA I-Hub without source tagging, hash, record_source, batch and trilha of source.
Persona	Data Engineer, Data Steward, QA Engineer and Analyst that investiga errors of ingestion.

2. Entities Bronze mandatory

Table	Purpose
bronze.raw_ingestion_batch	One row per ingestion run with orchestration metadata, job parameters and run status.
bronze.raw_source_file	One row per physical file/API response/web extraction artifact, including source tags.
bronze.raw_file_version	File version/hashes, raw path, effective timestamps and duplication/idempotency status.
bronze.raw_excel_sheet	Workbook sheet inventory with dimensions, detected header rows and parsing strategy.
bronze.raw_pdf_page	PDF page metadata, extraction/OCR confidence, page hash and section labels.
bronze.raw_extracted_row	Raw row-level records, preserving original row index, row hash and extraction warnings.
bronze.raw_extracted_cell	Cell-level lineage for Excel/PDF tables when granular traceability is required.
bronze.raw_indicator_observation	Main raw fact-like table: one extracted indicator observation with raw value, period, source and provenance.
bronze.raw_parse_event	Parsing warnings/errors, layout drift, OCR issues and detected schema changes.
bronze.raw_quarantine	Rejected or ambiguous records that must not be promoted automatically to Silver.

3. Mandatory fields from the source tagging

Field	Mandatory	Functional rule
source_institution	Yes	Institution of source: INE, BNA, MINFIN, UGD, BODIVA, CMC, ARSEG, ANPG etc.
source_tier	Yes	Tier 1, Tier 2 or Tier 3 according to catalog from the sources.
source_publication	Yes	Name of the publication/base/file original.
source_url_or_reference	Yes when available	Link official, path in the DateSources.zip or identificador of collection.
collection_method	Yes	API, download manual, web scraping, PDF extraction, Excel parsing or upload controlled.
extraction_timestamp	Yes	Timestamp UTC of the collection or parsing.
file_hash_sha256	Yes for files	Usado for idempotency and duplication detection.
module/domain/access_tier	Yes when identifiable	M1/M2/M3, domain and Base/Extended/Detail.
record_source	Yes	Identificador technical from the source in the Data Vault.

4. Pipeline functional Bronze

- Receive the file/API response and create raw_ingestion_batch.
- Calculate the SHA-256 hash of the artifact and verify idempotency.
- Classify source, module, domain, access tier, expected periodicity and collection method.
- Persist the raw file in storage and metadata in raw_source_file/raw_file_version.
- Detect structure: sheets, PDF pages, tables, rows, columns, headers and cells.
- Populate raw_indicator_observation with raw_value_text, raw_value_numeric when parseable, raw period, raw unit and row/cell/page reference.
- Record parse_event for layout drift, OCR confidence, missing columns, invalid values or unexpected headers.
- Send ambiguous records to raw_quarantine; never delete the raw file.
- Publish batch status and ingestion metrics to the observability layer.

5. Allowed Bronze Validations

Check	Severity	Action
File hash duplication	Info/Critical	If same hash already existe, not duplication observations active; record rerun.
Mandatory metadata missing	Critical	Quarantine of the batch or records up to preenchimento.
Numeric parse errorr	Warning	Gravar raw_value_text, make raw_value_numeric null and record flag.
Missing period	Critical	Block automatic promotion to Silver.
Duplicate observation key	Warning/Critical	Record duplication by indicator_candidate + source + period + file + row/cell.
Unit/scale drift	Warning	Flag change of unit/scale without correct.
Freshness warning	Warning	Calculate only the basic delay; formal status is calculated in the Silver BV.

6. Acceptance criteria

The Validation indicator configuration must cover the 45 eligible indicators and keep explicit versioning for new indicators, series, sources, periodicities and parsing rules.

- Each received file must have a hash and raw path, source_institution, source_tier, source_publication, collection_method and ingestion_batch_id.
- Reprocessing the same file with the same hash must not create an active duplication.
- Each raw_indicator_observation must point to file, version, sheet/page and, when possible, row/cell.
- Critical failures must go to quarantine and prevent automatic promotion to Silver.
- The pipeline must generate metrics: total rows read, observations created, parse errors, quarantined records and batch duration.
- Bronze must provide sufficient input to create hubs, links and satellites in the Silver Raw Vault.

2. Catalog and functional specification of the 45 indicators

The sections below consolidate the catalog and functional rules for the 45 eligible indicators. Previous DoD content, the single storage standard and common rules are now handled within the Bronze standard and the detailed specifications by indicator.

7. Summary catalog of the 45 included indicators

The indicators below are the 45 eligible indicators from the Validation sheet, The Logical subject column replaces the previous interpretation of one physical table per indicator.

#	Indicator	Module/ Tier	Source	Format	Periodicity	Physical Bronze target	Logical subject / optional view
1	Total GDP — real and nominal	M1.1 / Base	INE	XLSX	Quarterly	bronze.raw_indicator_observation	brz_m1_1_total_gdp_real_and_nominal
2	Oil vs. non-oil GDP	M1.1 / Base	INE	XLSX	Quarterly	bronze.raw_indicator_observation	brz_m1_1_oil_vs_non_oil_gdp
3	Detailed sector GDP / GVA by economic activity	M1.2 / Extended	INE	XLSX	Quarterly	bronze.raw_indicator_observation	brz_m1_2_detailed_sectoral_gdp_gva_by_economic_ac
4	National accounts by production, expenditure and income approach	M1.1 / Base	INE	XLSX	Annual	bronze.raw_indicator_observation	brz_m1_1_national_accounts_by_production_expendit
5	Industrial Production Index — overall index	M1.1 / Base	INE	XLSX	Monthly	bronze.raw_indicator_observation	brz_m1_1_industrial_production_index_overall_inde
6	Business tendency and confidence surveys — ICE headline	M1.1 / Base	INE	XLSX	Monthly	bronze.raw_indicator_observation	brz_m1_1_business_tendency_and_confidence_surveys
7	Oil, gas, condensate and mineral production	M1.1 / Base	ANPG	WEB SCRAPING	Monthly	bronze.raw_indicator_observation	brz_m1_1_oil_gas_condensate_and_mineral_productio
8	National CPI and	M1.1 / Base	INE	XLSX	Monthly	bronze.raw_indicator_observati	brz_m1_1_national_cpi_and_luanda_cpi_index_and_he

LAZA Bronze Raw Ingestion

#	Indicator	Module/ Tier	Source	Format	Periodicity	Physical Bronze target	Logical subject / optional view
	Luanda CPI — index and headline changes					on	
9	CPI by province — headlines	M1.2 / Extended	To be confirmed	To be confirmed	To be confirmed	bronze.raw_indicator_observation	brz_m1_2_cpi_by_province_headlines
10	Interest rates	M1.1 / Base	BNA	XLSX	Monthly	bronze.raw_indicator_observation	brz_m1_1_interest_rates
11	USD/AOA and EUR/AOA exchange rate — BNA reference	M1.1 / Base	BNA	Excel extraction	Daily	bronze.raw_indicator_observation	brz_m1_1_usd_aoa_and_eur_aoa_exchange_rate_bna_re
12	Balance of Payments — major items	M1.1 / Base	BNA	XLSX	Quarterly	bronze.raw_indicator_observation	brz_m1_1_balance_of_payments_major_items
13	International Investment Position — analytical	M1.1 / Base	BNA	XLSX	Quarterly	bronze.raw_indicator_observation	brz_m1_1_international_investment_position_analyt
14	Fiscal Macro Framework / General State Account - QMF table presentation	M1.1 / Base	MINFIN	XLSX	Annual	bronze.raw_indicator_observation	brz_m1_1_fiscal_macro_framework_general_state_acc
15	Public Finance - Government operations	M1.2 / Extended	To be confirmed	To be confirmed	Quarterly	bronze.raw_indicator_observation	brz_m1_2_public_finance_government_operations
16	Total population and main demographic indicators	M1.1 / Base	INE	XLSX	Annual / Decennial	bronze.raw_indicator_observation	brz_m1_1_total_population_and_main_demographic_in
17	Unemployment rate, employment and active population	M1.1 / Base	INE	XLSX	Quarterly / Annual	bronze.raw_indicator_observation	brz_m1_1_unemployment_rate_employment_and_active
18	Financial Soundness and Banking Sector Performance Indicators	M2.1 / Base	BNA	PDF	Quarterly / Annual	bronze.raw_indicator_observation	brz_m2_1_financial_soundness_and_banking_sector_p
19	Listed shares — prices, volumes and changes	M3.1 / Base	BODIVA	PDF	Daily	bronze.raw_indicator_observation	brz_m3_1_listed_shares_prices_volumes_and_changes
20	Provincial / regional GDP and GDP per capita	M1.2 / Extended	INE	XLSX	Annual / irregular	bronze.raw_indicator_observation	brz_m1_2_provincel_regional_gdp_and_gdp_per_capi
21	Construction Materials Price Index (IPMC)	M1.2 / Extended	INE	XLSX	Monthly	bronze.raw_indicator_observation	brz_m1_2_construction_materials_price_index_ipmc
22	Producer Price Index (IPP)	M1.2 / Extended	INE	XLSX	Monthly	bronze.raw_indicator_observation	brz_m1_2_producer_price_index_ipp
23	Wholesale Price Index (IPG)	M1.2 / Extended	INE	XLSX	Monthly	bronze.raw_indicator_observation	brz_m1_2_wholesale_price_index_ipg
24	Export/import price indices and terms of trade (IPEI)	M1.2 / Extended	INE	XLSX	Annual / as published	bronze.raw_indicator_observation	brz_m1_2_exposet_import_price_indices_and_terms_of
25	LUIBOR - all interbank maturities	M1.2 / Extended	BNA	XLSX	monthly	bronze.raw_indicator_observation	brz_m1_2_luibor_all_interbank_maturities
26	Primary market	M1.2 / Extended	BNA	XLSX	monthly	bronze.raw_indicator_observation	brz_m1_2_primary_market
2	Money market operations: OMAs,	M1.2 /	BNA	WEB SCRAPIN	Monthly	bronze.raw_indicator_observati	brz_m1_2_money_market_operations_omas_facilities_repos_and

LAZA Bronze Raw Ingestion

#	Indicator	Module/ Tier	Source	Format	Periodicity	Physical Bronze target	Logical subject / optional view
7	facilities, repos and interbank market; primary securities market	Extended		G		on	
28	Monetary aggregates and monetary base (M1, M2, M3 and counterparts)	M1.2 / Extended	BNA	XLSX	Monthly	bronze.raw_indicator_observation	brz_m1_2_monetary_aggregates_and_monetary_base_m1
29	BNA, OSD, OSF balance sheet and monetary surveys	M1.2 / Extended	BNA	XLSX	Monthly	bronze.raw_indicator_observation	brz_m1_2_bna_osd_osf_balance_sheet_and_monetary_s
30	Credit to the economy and credit by economic activity sector	M1.2 / Extended	BNA	XLSX	Monthly	bronze.raw_indicator_observation	brz_m1_2_credit_to_the_economy_and_credit_by_econ
31	Deposits and loans by institutional sector, currency and maturity	M1.2 / Extended	BNA	XLSX	Monthly	bronze.raw_indicator_observation	brz_m1_2_deposits_and_loans_by_institutional_sect
32	Bank rates by maturity, currency, product and segment	M1.2 / Extended	BNA	XLSX	Monthly	bronze.raw_indicator_observation	brz_m1_2_bank_rates_by_maturity_currency_product_and_segmen
33	Reserves template: official reserve assets, instrument and currency	M1.2 / Extended	BNA	XLSX	Monthly	bronze.raw_indicator_observation	brz_m1_2_reserves_template_official_reserve_asset
34	Balance of Payments — complete analytical and standard presentation	M1.2 / Extended	BNA	XLSX	Quarterly	bronze.raw_indicator_observation	brz_m1_2_balance_of_payments_complete_analytical
35	Exports by product/destination	M1.2 / Extended	BNA	XLSX	Quarterly / Annual	bronze.raw_indicator_observation	brz_m1_2_exports_by_product_destination
36	Imports by economic category, product and origin	M1.2 / Extended	BNA	XLSX	Quarterly / Annual	bronze.raw_indicator_observation	brz_m1_2_imports_by_economic_category_product_and
37	Main External Indicators	M1.2 / Extended	BNA	XLSX/ PDF	Quarterly / Annual	bronze.raw_indicator_observation	brz_m1_2_main_externall_indicators
38	FDI detail, by sector	M1.2 / Extended	BNA	XLSX	Quarterly / Annual	bronze.raw_indicator_observation	brz_m1_2_fdi_detail_by_sector
39	Public debt evolution — domestic/externall stock, currency, maturity and instrument - Overview	M1.2 / Extended	MINFIN; UGD	PDF	Quarterly	bronze.raw_indicator_observation	brz_m1_2_public_debt_evolution_domestic_externall
40	Main Activity Indicators	M2.2 / Extended	Commercial Banks	XLSX/ PDF	Quarterly / Annual	bronze.raw_indicator_observation	brz_m2_2_main_activity_indicators
41	Balance Sheet and Income Statement by Bank (summary)	M2.3 / Detail	BNA; Commercial banks	XLSX/ PDF	Quarterly / Annual	bronze.raw_indicator_observation	brz_m2_3_balance_sheet_and_income_statement_by_bank_summary
42	Domestic sovereign yield curve	M3.2 / Extended	UGD	PDF	Weekly / Monthly	bronze.raw_indicator_observation	brz_m3_2_domestic_sovereign_yield_curve
4	Corporate bonds and commercial	M3.3 /	BODIVA;	PDF/WEB	whenever	bronze.raw_indicator_observati	brz_m3_3_corporate_bonds_and_commercial_paper_ipo

#	Indicator	Module/ Tier	Source	Format	Periodicity	Physical Bronze target	Logical subject / optional view
3	paper/ IPO	Detail	CMC		updated	on	
44	Economic Diversification Index	M4.2 / Extended	Laza	Indicator calculated	Semiannual	bronze.raw_indicator_observation	brz_m4_2_economic_diversification_index
45	Balance Sheet Center by sector of activity	M1.3 / Detail	BNA	XLSX	Annual	bronze.raw_indicator_observation	brz_m1_3_balance_sheet_center_by_sector_of_activi

8. Functional specification detailed by indicator

Each following section should be read as a functional development backlog item. The implementation standard is the same for all indicators: metadata and files in source/batch/version entities, physical source data in sheets/pages/rows/cells, and candidate values in the central table bronze.raw_indicator_observation.

Indicator 01 - Total GDP — real and nominal

Source	INE
Format	XLSX
Periodicity	Quarterly
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_total_gdp_real_and_nominal
Sheet/section/columns to read	GROSS DOMESTIC PRODUCT = (GDP oil (Petroleum Extraction and Refining) column C+ | GDP NON-OIL Column D); Column T
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of total nominal and real GDP based on quarterly national accounts.

Objective: Make main measure of economic performance available in Bronze, preserving traceability to the official source.

Persona: Macroeconomic analyst / executive user of the cockpit M1

2 Development scope

In scope

- Configure the indicator 01 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: GROSS DOMESTIC PRODUCT = (GDP oil (Petroleum Extraction and Refining) column C+ | GDP NON-OIL Column D); Column T
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Total GDP — real and nominal into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 01, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 01 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='01' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: GROSS DOMESTIC PRODUCT = (GDP oil (Petroleum Extraction and Refining) column C+ | GDP NON-OIL Column D)
- Validation spreadsheet rule: Column T
- Expected periodicity: Quarterly; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.
- Functional subcomponents/line items to observe: Nominal; Real.

Formulas

- Silver/Gold: GDP nominal total = GDP Oil (Q1 column C) + GDP Non-oil (Q1 column D). Real GDP total: use the indicated reading in Q2, column T. In Bronze, persist raw values and columns of source without recalculate.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.

- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 02 - Oil vs. non-oil GDP

Source	INE
Format	XLSX
Periodicity	Quarterly
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_oil_vs_non_oil_gdp
Sheet/section/columns to read	GDP oil (Petroleum Extraction and Refining) column C (Q1); Column E; GDP NON-OIL (Column D); Column T - Column E
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of nominal and real oil and non-oil GDP components.

Objective: Enable separate analysis of the oil and non-oil economy based on the official source.

Persona: Macro economist / economic diversification analyst

2 Development scope

In scope

- Configure the indicator 02 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: GDP oil (Petroleum Extraction and Refining) column C (Q1); Column E; GDP NON-OIL (Column D); Column T - Column E
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Oil vs. non-oil GDP into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 02, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 02 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='02' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: GDP oil (Petroleum Extraction and Refining) column C (Q1)
- Validation spreadsheet rule: Column E
- Validation spreadsheet rule: GDP NON-OIL (Column D)
- Validation spreadsheet rule: Column T - Column E
- Expected periodicity: Quarterly; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.
- Functional subcomponents/line items to observe: Oil / Nominal; Real; Non-oil / Nominal.

Formulas

- Silver/Gold: Oil nominal = Q1 column C; Oil real = Q2 column E; Non-oil nominal = Q1 column D; Non-oil real = Q2 column T - Q2 column E. Bronze: load C, D, E and T as raw values with column metadata.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 03 - Detailed sector GDP / GVA by economic activity

Source	INE
Format	XLSX
Periodicity	Quarterly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_detailed_sectoral_gdp_gva_by_economic_ac
Sheet/section/columns to read	All Q2 columns except column T
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the GDP/GVA detailed by economic activity.

Objective: Make the sector composition of the economy available for later modeling in Silver/Gold.

Persona: Sector analyst / economic consultant

2 Development scope

In scope

- Configure the indicator 03 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: all Q2 columns except column T
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Detailed sector GDP / GVA by economic activity into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 03, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 03 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='03' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.

- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All Q2 columns except column T
- Expected periodicity: Quarterly; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Do not calculate totals in Bronze. Load all Q2 columns except column T, according to the validation rule.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 04 - National accounts by production, expenditure and income approach

Source	INE
Format	XLSX
Periodicity	Annual
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_national_accounts_by_production_expendit
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of annual national accounts by production, expenditure and income approaches.

Objective: Create a structured annual base for comparative analysis between national accounts approaches.

Persona: National accounts analyst / M1 product team

2 Development scope

In scope

- Configure the indicator 04 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator National accounts by production, expenditure and income approach into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 04, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 04 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='04' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
- Expected periodicity: Annual; update SLA: Up to 15 business days after official publication; validate the reference year.
- Expected period fields: reference_year.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Bronze must not recalculate aggregates. Preserve the positive and negative signs from the source; total rows and calculated rows are not included in the main observation table.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 05 - Industrial Production Index — overall index

Source	INE
Format	XLSX
Periodicity	Monthly
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_industrial_production_index_overall_inde
Sheet/section/columns to read	Only Table 1 - IPI (Industrial Production Index 2010 = 100) (removing the rows of totals and calculated) Pay attention to signs (+ and -)
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the Index of overall industrial production.

Objective: Allow monthly monitoring of industrial production as a leading indicator of the real economy.

Persona: Real activity analyst / M1 Base user

2 Development scope

In scope

- Configure the indicator 05 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: load only Table 1 - IPI (Industrial Production Index 2010 = 100), removing total and calculated rows. Pay attention to signs (+ and -).
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Industrial Production Index — overall index into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 05, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 05 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='05' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load only Table 1 - IPI (Industrial Production Index 2010 = 100), removing total and calculated rows. Pay attention to signs (+ and -).
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load only Table 1 — IPI (2010=100). Do not recalculate the index in Bronze; variations will be derived in Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 06 - Business tendency and confidence surveys — ICE headline

Source	INE
Format	XLSX
Periodicity	Monthly
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_business_tendency_and_confidence_surveys
Sheet/section/columns to read	Load the entire spreadsheet (as is)
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the indicator ICE of tendency and confidence corporate.

Objective: Make available economic sentiment indicator in Bronze for analyses of economic outlook.

Persona: Economic outlook analyst / leading indicators user

2 Development scope

In scope

- Configure the indicator 06 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load the entire spreadsheet (as is)
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Business tendency and confidence surveys — ICE headline into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 06, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 06 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='06' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.

- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load the entire spreadsheet (as is)
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load the ICE spreadsheet exactly as published. Average, variation and normalization calculations remain outside Bronze.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 07 - Oil, gas, condensate and mineral production

Source	ANPG
Format	WEB SCRAPING
Periodicity	Monthly
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_oil_gas_condensate_and_mineral_productio
Sheet/section/columns to read	Load the data of the page extraindo the indicators
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion or scraping of the main indicators of production oil, gas, ALNG, LPG and sondagem.

Objective: Cover the extractive sector, which is essential for GDP, exports and public revenues.

Persona: Oil, gas and extractive industries analyst / M1 Base user

2 Development scope

In scope

- Configure the indicator 07 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load the data of the page extraindo the indicators
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Oil, gas, condensate and mineral production into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 07, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 07 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='07' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.

- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load the data of the page extraindo the indicators
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.
- Subcomponents/line items functional to observe: production of petroleum; average daily; production of gas associated; average daily gas; factory ALNG; average daily of ALNG; Cabinda, the production of LPG; units of sondagem.

Formulas

- Bronze must persist metrics by product/period exactly the extracted. Daily averages or already published indicators must be marked as “source_calculated”; if the average is internally derived, calculate it only in Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.
- Web page/API unavailable or blocked: record SOURCE_UNAVAILABLE, save a snapshot when possible and apply controlled retry.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 08 - National CPI and Luanda CPI — index and headline changes

Source	INE
Format	XLSX
Periodicity	Monthly
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_national_cpi_and_luanda_cpi_index_and_he
Sheet/section/columns to read	Load all spreadsheets in the file, with the first being general and the others by province | Load only Index and Contribution data | Exclude Variation (it must be calculated as a variable)
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the National CPI and Luanda CPI, including index and contribution.

Objective: Make the inflation headline and its components basic for analysis monthly.

Persona: Inflation analyst / economic decision-maker

2 Development scope

In scope

- Configure the indicator 08 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load all spreadsheets in the file, with the first being general and the others by province | Load only Index and Contribution data | Exclude Variation (it must be calculated as a variable)
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator National CPI and Luanda CPI — index and headline changes into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 08, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 08 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='08' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load all spreadsheets in the file, with the first being general and the others by province | Load only Index and Contribution data | Exclude Variation (it must be calculated as a variable)
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load only Index and Contribution. Exclude Variation from the main load; in Silver/Gold calculate MoM = index_t / index_t-1 - 1 and YoY = index_t / index_t-12 - 1. In Bronze, persist only raw values and metadata; final calculations belong to Silver/Gold layers.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 09 - CPI by province — headlines

Source	To be confirmed
Format	XLSX
Periodicity	To be confirmed
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_cpi_by_province_headlines
Sheet/section/columns to read	Load all spreadsheets in the file, with the first being general and the others by province | Load only Index and Contribution data | Exclude Variation (it must be calculated as a variable)
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the series of inflation by province.

Objective: Allow territorial analysis of inflationary pressures.

Persona: Regional analyst / provincel inflation analyst

2 Development scope

In scope

- Configure the indicator 09 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load all spreadsheets in the file, with the first being general and the others by province | Load only Index and Contribution data | Exclude Variation (it must be calculated as a variable)
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator CPI by province — headlines into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 09, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 09 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='09' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load all spreadsheets in the file, with the first being general and the others by province | Load only Index and Contribution data | Exclude Variation (it must be calculated as a variable)
- Expected periodicity: to be confirmed; update SLA: according to the periodicity published by the source; record publication date and extraction date.
- Expected period fields: reference_period, reference_date when available.
- Classify the source as Tier 3 or N/A — validate provenance and keep this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load only Index and Contribution by province when available. Variations must be derived later using the same logic as the National CPI. In Bronze, persist only raw values and metadata; final calculations belong to Silver/Gold layers.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 10 - Interest rates

Source	BNA
Format	XLSX
Periodicity	Monthly
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_interest_rates
Sheet/section/columns to read	Load all data from the spreadsheet | Except rate LUIBOR
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the interest rates of the Bank Central, excluding LUIBOR.

Objective: Create base of reference for monetary policy without duplication curves interbanking.

Persona: Monetary policy analyst / user M1 Base

2 Development scope

In scope

- Configure the indicator 10 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load all data from the spreadsheet | Except rate LUIBOR
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Interest rates into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 10, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 10 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='10' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.

- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load all data from the spreadsheet | Except rate LUIBOR
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load all the rates of the spreadsheet except LUIBOR. Do not calculate spreads in Bronze.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 11 - USD/AOA and EUR/AOA exchange rate — BNA reference

Source	BNA
Format	Excel extraction
Periodicity	Daily
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_usd_aoa_and_eur_aoa_exchange_rate_bna_re
Sheet/section/columns to read	Perform daily extraction of currency quotes (USD and EUR) | Bring the history in the first load (history from 2015 onward)
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion daily of the USD/AOA and EUR/AOA rates of reference of the BNA.

Objective: Keep history foreign-exchange daily desde 2015 and load incremental daily.

Persona: Foreign-exchange analyst / treasury / markets user

2 Development scope

In scope

- Configure the indicator 11 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Perform daily extraction of currency quotes (USD and EUR) | Bring the history in the first load (history from 2015 onward)
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator USD/AOA and EUR/AOA exchange rate — BNA reference into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 11, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 11 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='11' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Perform daily extraction of currency quotes (USD and EUR) | Bring the history in the first load (history from 2015 onward)
- Expected periodicity: Daily; update SLA: Up to D+1 after official publication; ingestion incremental daily.
- Expected period fields: reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Bronze stores raw quotations. Daily variations, monthly averages and accumulated returns remain in Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 12 - Balance of Payments — major items

Source	BNA
Format	XLSX
Periodicity	Quarterly
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_balance_of_payments_major_items
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the main items of the Balance of Payments.

Objective: Make externall balances and components available for macroeconomic reading.

Persona: External sector analyst / macro research

2 Development scope

In scope

- Configure the indicator 12 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Balance of Payments — major items into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 12, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 12 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='12' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Expected periodicity: Quarterly; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Preserve signs from the source. Do not load rows in bold, subtotals or calculated rows as main observations; identities of the BoP will be validated in Silver/Gold. In Bronze, persist only raw values and metadata; final calculations belong to Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Cross-Series Consistency: run informative validation of accounting/fiscal identities when published totals exist; flag only.

Indicator 13 - International Investment Position — analytical

Source	BNA
Format	XLSX
Periodicity	Quarterly
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_international_investment_position_analyt
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the analytical International Investment Position.

Objective: Allow analysis of stocks of externall assets and liabilities.

Persona: External position analyst / country risk analyst

2 Development scope

In scope

- Configure the indicator 13 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator International Investment Position — analytical into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 13, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 13 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='13' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Expected periodicity: Quarterly; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Preserve signs and units. Do not recalculate position net in Bronze; load components of assets and liabilities according to source.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Cross-Series Consistency: run informative validation of accounting/fiscal identities when published totals exist; flag only.

Indicator 14 - Fiscal Macro Framework / General State Account - QMF table presentation

Source	MINFIN
Format	XLSX
Periodicity	Annual
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_fiscal_macro_framework_general_state_acc
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the macro-fiscal framework / General State Account in quarterly series.

Objective: Make available official-source revenues, expenditures and fiscal balances.

Persona: Tax analyst / M1 Base user

2 Development scope

In scope

- Configure the indicator 14 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Fiscal Macro Framework / General State Account - QMF table presentation into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 14, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 14 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='14' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Expected periodicity: Annual; update SLA: Up to 15 business days after official publication; validate the reference year.
- Expected period fields: reference_year.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Do not calculate fiscal aggregates in Bronze. Exclude calculated totals/subtotals and preserve source signs and line items.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Cross-Series Consistency: run informative validation of accounting/fiscal identities when published totals exist; flag only.

Indicator 15 - Public Finance - Government operations

Source	To be confirmed
Format	XLSX
Periodicity	Quarterly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_public_finance_government_operations
Sheet/section/columns to read	A reading must seguir the rule of source defined in the sheet Validation and in the file source correspondente.
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the operations of finance public based on the series quarterly.

Objective: Complement the fiscal framework with operational granularity for government analysis.

Persona: Tax analyst / government operations analyst

2 Development scope

In scope

- Configure the indicator 15 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: The reading must seguir the rule of source defined in the sheet Validation and in the file source correspondente.
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Public Finance - Government operations into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 15, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 15 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='15' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- There is no detailed specific rule in the Validation sheet; apply the standard Bronze ingestion rule and validate with the Product Owner before promotion to Silver.
- Expected periodicity: Quarterly; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 3 or N/A — validate provenance and keep this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Use the same inferred file as indicator 14. Balance, ratio and execution calculations remain outside Bronze.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 16 - Total population and main demographic indicators

Source	INE
Format	XLSX
Periodicity	Annual / Decennial
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_total_population_and_main_demographic_in
Sheet/section/columns to read	Load the table toda
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the population total and indicators demographic main.

Objective: Create dimension populacional for analyses per capita and segmentation socioeconomic.

Persona: Demographics analyst / market economist

2 Development scope

In scope

- Configure the indicator 16 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load the table toda
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Total population and main demographic indicators into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 16, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 16 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='16' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.

- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load the table toda
- Expected periodicity: Annual / Decennial; update SLA: Up to 15 business days after official publication; validate the reference year.
- Expected period fields: reference_year.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load table complete. Indicators per capita will be calculated in Silver/Gold with combination of population and metrics economics. In Bronze, persist only raw values and metadata; final calculations belong to Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 17 - Unemployment rate, employment and active population

Source	INE
Format	XLSX
Periodicity	Quarterly / Annual
Module/Tier	M1.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_1_unemployment_rate_employment_and_active
Sheet/section/columns to read	Load the table toda - the is historical the outra is the value most current - (Attention to the namemclaturas of names)
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of unemployment, employment and population active in methodologies old and new.

Objective: Keep comparable labor history, preserving the source methodology of each series.

Persona: Labor analyst / investigador social

2 Development scope

In scope

- Configure the indicator 17 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load the table toda - the is historical the outra is the value most current - (Attention to the namemclaturas of names)
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Unemployment rate, employment and active population into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 17, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 17 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='17' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load the table toda - the is historical the outra is the value most current - (Attention to the namemclaturas of names)
- Expected periodicity: Quarterly / Annual; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Bronze must load both published series and mark methodology_version = old_13_CIET or new_19_20_21_CIET. Harmonization between methodologies remains outside Bronze.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 18 - Financial Soundness and Banking Sector Performance Indicators

Source	BNA
Format	PDF
Periodicity	Quarterly / Annual
Module/Tier	M2.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m2_1_financial_soundness_and_banking_sector_p
Sheet/section/columns to read	Download the PDF, load the table from the PDF (single sheet), download all PDFs and load those that follow the same standard as the latest one
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the financial soundness and banking performance indicators published in PDF.

Objective: Make available structured banking KPIs with provenance and quality.

Persona: Financial stability analyst / M2 team

2 Development scope

In scope

- Configure the indicator 18 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Download the PDF, load the table from the PDF (single sheet), download all PDFs and load those that follow the same standard as the latest one
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Financial Soundness and Banking Sector Performance Indicators into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 18, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 18 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='18' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Download the PDF, load the table from the PDF (single sheet), download all PDFs and load those that follow the same standard as the latest one
- Expected periodicity: Quarterly / Annual; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Extract table single from the PDF. Do not recalculate ratios in Bronze; keep names PT/EN when available.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.
- PDF without table extractable or with low confidence: record PDF_EXTRACTION_LOW_CONFIDENCE and encaminhar for validation manual.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 19 - Listed shares — prices, volumes and changes

Source	BODIVA
Format	PDF
Periodicity	Daily
Module/Tier	M3.1 / Base
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m3_1_listed_shares_prices_volumes_and_changes
Sheet/section/columns to read	download each daily bulletin and load the tables that are part of chapter 1.2
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion daily of actions listed, prices, volumes, variations and debt public traded in the bulletin BODIVA.

Objective: Make available market secondary and instruments public in the module Corporate & Markets.

Persona: Equity markets analyst / M3 user

2 Development scope

In scope

- Configure the indicator 19 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: download each daily bulletin and load the tables that are part of chapter 1.2
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Listed shares — prices, volumes and changes into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 19, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 19 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='19' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: download each daily bulletin and load the tables that are part of chapter 1.2
- Expected periodicity: Daily; update SLA: Up to D+1 after official publication; ingestion incremental daily.
- Expected period fields: reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Bronze must extract tables from chapter 1.2. Variations, accumulated liquidity and proprietary indexes remain outside Bronze.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.
- PDF without table extractable or with low confidence: record PDF_EXTRACTION_LOW_CONFIDENCE and encaminhar for validation manual.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 20 - Provincial / regional GDP and GDP per capita

Source	INE
Format	XLSX
Periodicity	Annual / irregular
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_provincel_regional_gdp_and_gdp_per_capi
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the GDP provincel and GDP per capita.

Objective: Allow analysis regional of economic activity and comparison between provinces.

Persona: Regional analyst / territorial planning analyst

2 Development scope

In scope

- Configure the indicator 20 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Provincial / regional GDP and GDP per capita into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 20, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 20 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='20' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
- Expected periodicity: Annual / irregular; update SLA: Up to 15 business days after official publication; validate the reference year.
- Expected period fields: reference_year.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load Q1 and Q3, excluding totals and calculated. Published GDP per capita must be marked as source_calculated if it already comes from the source. In Bronze, persist only raw values and metadata; final calculations belong to Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 21 - Construction Materials Price Index (IPMC)

Source	INE
Format	XLSX
Periodicity	Monthly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_construction_materials_price_index_ipmc
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the Construction Materials Price Index.

Objective: Capture cost pressures in construction and infraestrutura.

Persona: Construction / sector inflation analyst

2 Development scope

In scope

- Configure the indicator 21 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Construction Materials Price Index (IPMC) into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 21, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 21 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='21' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Do not recalculate the index in Bronze. Load line items and periods of the sheet Index, excluding aggregates calculated.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 22 - Producer Price Index (IPP)

Source	INE
Format	XLSX
Periodicity	Monthly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_producer_price_index_ipp
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the Producer Price Index.

Objective: Anticipate cost pressures before they are reflected in consumer prices.

Persona: Producer prices / industry analyst

2 Development scope

In scope

- Configure the indicator 22 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Producer Price Index (IPP) into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 22, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 22 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='22' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load sheet Index the raw observations; variations monthly and year-on-year derived remain to Silver/Gold. In Bronze, persist only raw values and metadata; final calculations belong to Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 23 - Wholesale Price Index (IPG)

Source	INE
Format	XLSX
Periodicity	Monthly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_wholesale_price_index_ipg
Sheet/section/columns to read	load the columns A, B and C for the tres sheets
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the IPG global, importado and national.

Objective: Separate the dynamics of wholesale prices between domestic and imported components.

Persona: Wholesale inflation / trade analyst

2 Development scope

In scope

- Configure the indicator 23 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: load the columns A, B and C for the tres sheets
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Wholesale Price Index (IPG) into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 23, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 23 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='23' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.

- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: load the columns A, B and C for the tres sheets
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load columns A, B and C in the three indicated sheets. Do not calculate additional aggregations in Bronze.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 24 - Export/import price indices and terms of trade (IPEI)

Source	INE
Format	XLSX
Periodicity	Annual / as published
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_exposet_import_price_indices_and_terms_of
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the indexes of prices of export/imports and terms of troca.

Objective: Make the components necessary to the analysis of competitividade externall.

Persona: Foreign trade / terms of trade analyst

2 Development scope

In scope

- Configure the indicator 24 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Export/import price indices and terms of trade (IPEI) into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 24, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 24 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='24' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
- Expected periodicity: Annual / as published; update SLA: Up to 15 business days after official publication; validate the reference year.
- Expected period fields: reference_year.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Bronze loads published indexes and terms of trade. Subsequent formula: Terms of Trade = Export Price Index / Import Price Index * 100, when not published.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 25 - LUIBOR - all interbank maturities

Source	BNA
Format	XLSX
Periodicity	monthly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_luibor_all_interbank_maturities
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the LUIBOR by all the maturities interbanking.

Objective: Construir curve of reference of the interbank market for analysis of liquidity.

Persona: Money market / treasury analyst

2 Development scope

In scope

- Configure the indicator 25 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator LUIBOR - all interbank maturities into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 25, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 25 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='25' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -)
- Expected periodicity: monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load MMI and LUIBOR by maturity. Spreads between maturities will be derived outside Bronze.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 26 - Primary market

Source	BNA
Format	XLSX
Periodicity	monthly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_primary_market
Sheet/section/columns to read	Load the table Toda - EXCETO the columns TBC, OTIPC, OTTXC and OTBT
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the average interest rates of the securities in the primary market.

Objective: Monitor the cost of sovereign financing in primary issuances.

Persona: Debt / primary market analyst

2 Development scope

In scope

- Configure the indicator 26 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load the table Toda - EXCETO the columns TBC, OTIPC, OTTXC and OTBT
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Primary market into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 26, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 26 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='26' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load the table Toda - EXCETO the columns TBC, OTIPC, OTTXC and OTBT
- Expected periodicity: monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load the full table except columns TBC, OTIPC, OTTXC and OTBT. Curves and spreads remain outside Bronze.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 27 - Money market operations: OMAs, facilities, repos and interbank market; primary securities market

Source	BNA
Format	WEB SCRAPING
Periodicity	Monthly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_money_market_operations_omas_facilities_repos_and
Sheet/section/columns to read	Extract data from the applicable BNA page, preserving date, instrument, rate, amount, maturity, counterparty/segment when available and identification of the operation.
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the operations of the money market, including OMAs, facilities, repos, interbank market and primary market of securities.

Objective: Monitor liquidity, monetary intervention and operational dynamics of the market short-term financial market.

Persona: Monetary policy / treasury / money market analyst

2 Development scope

In scope

- Configure the indicator 27 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Extract data from the applicable BNA page, preserving date, instrument, rate, amount, maturity, counterparty/segment when available and identification of the operation.
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Money market operations: OMAs, facilities, repos and interbank market; primary securities market into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 27, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.

- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 27 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='27' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Extract data from the applicable BNA page, preserving date, instrument, rate, amount, maturity, counterparty/segment when available and identification of the operation.
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.
- Subcomponents/line items functional to observe: OMAs; standing facilities; repos; interbank market; primary market of securities.

Formulas

- Bronze must persist operations and published rates. Aggregations by instrument, weighted averages, spreads and indicators derived must be calculated only in Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP, connector, URL or collection folder: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.
- Web page/API unavailable or blocked: record SOURCE_UNAVAILABLE, save a snapshot when possible and apply controlled retry.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 28 - Monetary aggregates and monetary base (M1, M2, M3 and counterparts)

Source	BNA
Format	XLSX
Periodicity	Monthly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_monetary_aggregates_and_monetary_base_m1
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of aggregates monetary aggregates and base monetary.

Objective: Make available liquidity monetary and its components the base for analysis macro-financial.

Persona: Analyst monetary / economista M1

2 Development scope

In scope

- Configure the indicator 28 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Monetary aggregates and monetary base (M1, M2, M3 and counterparts) into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 28, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 28 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='28' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Do not recalculate M1/M2/M3 in Bronze. Load components and totals permitidos according to source, excluding subtotals calculated when indicatesdos.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 29 - BNA, OSD, OSF balance sheet and monetary surveys

Source	BNA
Format	XLSX
Periodicity	Monthly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_bna_osd_osf_balance_sheet_and_monetary_s
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the balance sheets of the BNA, OSD, OSF and surveys monetary aggregates.

Objective: Create the base raw of assets/liabilities for analysis monetary and intermediation financial.

Persona: Financial system analyst / M2 data engineer

2 Development scope

In scope

- Configure the indicator 29 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator BNA, OSD, OSF balance sheet and monetary surveys into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 29, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 29 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='29' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Preserve accounting structure and signs. Accounting-identity validations will be quality checks, not Bronze transformations.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Cross-Series Consistency: run informative validation of accounting/fiscal identities when published totals exist; flag only.

Indicator 30 - Credit to the economy and credit by economic activity sector

Source	BNA
Format	XLSX
Periodicity	Monthly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_credit_to_the_economy_and_credit_by_econ
Sheet/section/columns to read	Load both sheets
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of credit to the economy by sector of activity.

Objective: Monitor credit allocation and sector dynamism.

Persona: Credit / macro-financial analyst

2 Development scope

In scope

- Configure the indicator 30 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load both sheets
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Credit to the economy and credit by economic activity sector into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 30, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 30 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='30' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.

- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load both sheets
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load the two sheets CAE REV1 and CAE REV2. Harmonization of classifications between revisions is outside the Bronze scope.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 31 - Deposits and loans by institutional sector, currency and maturity

Source	BNA
Format	XLSX
Periodicity	Monthly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_deposits_and_loans_by_institutional_sect
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of deposits and loans by institutional sector, currency and maturity.

Objective: Make available granularity of financial intermediation for analysis by sector and currency.

Persona: Analyst banking / risk financial

2 Development scope

In scope

- Configure the indicator 31 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Deposits and loans by institutional sector, currency and maturity into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 31, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 31 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='31' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load loans and deposits in respective files/sheets. Do not calculate ratios credit/deposit in Bronze.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 32 - Bank rates by maturity, currency, product and segment

Source	BNA
Format	XLSX
Periodicity	Monthly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_bank_rates_by_maturity_currency_product_and_segmen
Sheet/section/columns to read	Load the file of export BNA for banking rates, preserving maturity, currency, product, segment, period and observed rate.
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the banking rates by maturity, currency, product and segment.

Objective: Make the view granular of the transmission of the rates for products banking, deposits and credit.

Persona: Banking analyst / monetary policy and credit analyst

2 Development scope

In scope

- Configure the indicator 32 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load the file of export BNA for banking rates, preserving maturity, currency, product, segment, period and observed rate.
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Bank rates by maturity, currency, product and segment into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 32, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 32 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='32' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load the file of export BNA for banking rates, preserving maturity, currency, product, segment, period and observed rate.
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.
- Subcomponents/line items functional to observe: maturity; currency; product; segment; observed rate.

Formulas

- Bronze must load the raw rates by dimension. Calculation of spreads, averages by product and curves by maturity must occur in Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP, connector, URL or collection folder: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 33 - Reserves template: official reserve assets, instrument and currency

Source	BNA
Format	XLSX

Periodicity	Monthly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_reserves_template_official_reserve_asset
Sheet/section/columns to read	Load the rows of 1 to 27 that are the most importantes
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the template of international ratings and liquidity in foreign currency.

Objective: Make available composition on assets of reserva official.

Persona: International ratings / externall sector analyst

2 Development scope

In scope

- Configure the indicator 33 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load the rows of 1 to 27 that are the most importantes
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Reserves template: official reserve assets, instrument and currency into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 33, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 33 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='33' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.

- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load the rows of 1 to 27 that are the most importantes
- Expected periodicity: Monthly; update SLA: up to 5 business days after official publication; validate the absence of monthly gaps.
- Expected period fields: reference_year, reference_month, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load rows 1 to 27 of the Angola sheet. Currency conversions and import-month coverage calculations remain outside Bronze.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 34 - Balance of Payments — complete analytical and standard presentation

Source	BNA
Format	XLSX
Periodicity	Quarterly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_balance_of_payments_complete_analytical
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion complete of the Balance of Payments in presentation analytical and standard.

Objective: Provide greater detail of external accounts for specialized analysis.

Persona: External sector analyst / BoP specialist

2 Development scope

In scope

- Configure the indicator 34 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Balance of Payments — complete analytical and standard presentation into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 34, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 34 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='34' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Expected periodicity: Quarterly; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Preserve signs, exclude calculated totals/subtotals and keep the hierarchy of line items for subsequent validation. In Bronze, persist only raw values and metadata; final calculations belong to Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Cross-Series Consistency: run informative validation of accounting/fiscal identities when published totals exist; flag only.

Indicator 35 - Exports by product/destination

Source	BNA
Format	XLSX
Periodicity	Quarterly / Annual
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_exports_by_product_destination
Sheet/section/columns to read	Load export tables by category/product and by country/destination. Exclude totals, subtotals and calculated fields when they exist.
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the exports by product and destination, with foco in the composition of the pauta exportedra and in the main markets externall.

Objective: Allow analysis of concentration, dependency and evolution of the revenues of export by product and country of destination.

Persona: Foreign trade / macroeconomic analyst

2 Development scope

In scope

- Configure the indicator 35 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load export tables by category/product and by country/destination. Exclude totals, subtotals and calculated fields when they exist.
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Exports by product/destination into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 35, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.

- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 35 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='35' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load export tables by category/product and by country/destination. Exclude totals, subtotals and calculated fields when they exist.
- Expected periodicity: Quarterly / Annual; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.
- Subcomponents/line items functional to observe: product; category; destination/country parceiro; value exported.

Formulas

- Bronze must persist values exporteds by product, category and destination according to published. Share percentage, rankings and concentration by destination will be derived in Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP, connector, URL or collection folder: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Cross-Series Consistency: run informative validation between series related when totals, ratios or components are published; flag only.

Indicator 36 - Imports by economic category, product and origin

Source	BNA
Format	XLSX
Periodicity	Quarterly / Annual
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_imports_by_economic_category_product_and
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of imports by category, classification economic, product, source and exports by category.

Objective: Create base of foreign trade for analysis of dependency importada and structure exportedra.

Persona: Foreign trade / imports and exports analyst

2 Development scope

In scope

- Configure the indicator 36 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Imports by economic category, product and origin into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 36, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 36 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='36' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Expected periodicity: Quarterly / Annual; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load detailed, non-calculated rows. Trade balances, shares and rankings remain in Silver/Gold. In Bronze, persist only raw values and metadata; final calculations belong to Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 37 - Main External Indicators

Source	BNA
Format	XLSX/PDF
Periodicity	Quarterly / Annual
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_main_externall_indicators
Sheet/section/columns to read	Extract data from the tables UGD/BNA applicable, keeping separation explicit between externall debt raw and debt public externall.
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the main externall indicators, separating raw externall debt, public externall debt and metrics associated with the externall sector.

Objective: Support analysis of externall vulnerability, sustainability and exposure to international financing.

Persona: External sector analyst / risk country / externall debt

2 Development scope

In scope

- Configure the indicator 37 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Extract data from the tables UGD/BNA applicable, keeping separation explicit between externall debt raw and debt public externall.
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Main External Indicators into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 37, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 37 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='37' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Extract data from the tables UGD/BNA applicable, keeping separation explicit between externall debt raw and debt public externall.
- Expected periodicity: Quarterly / Annual; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.
- Subcomponents/line items functional to observe: externall debt raw; debt public externall; indicators externall; ratios published when exist.

Formulas

- Bronze must keep the conceitos as published. Ratios, reconciliations and indicators composites will be handleds in the layers Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP, connector, URL or collection folder: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.
- PDF without table extractable or with low confidence: record PDF_EXTRACTION_LOW_CONFIDENCE and encaminhar for validation manual.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Cross-Series Consistency: run informative validation between series related when totals, ratios or components are published; flag only.

Indicator 38 - FDI detail, by sector

Source	BNA
Format	XLSX
Periodicity	Quarterly / Annual
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_fdi_detail_by_sector
Sheet/section/columns to read	All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of FDI by sector.

Objective: Make available sector attraction of foreign capital and its temporal evolution.

Persona: External investment / sector analyst

2 Development scope

In scope

- Configure the indicator 38 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator FDI detail, by sector into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 38, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 38 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='38' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: All columns and rows (removing total and calculated rows). Pay attention to signs (+ and -) | Verify rows in bold - are calculated rows do not need to be loaded | Subtotal rows also must not be loaded
- Expected periodicity: Quarterly / Annual; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Load raw sector observations, exclude calculated subtotals and preserve signs. In Bronze, persist only raw values and metadata; final calculations belong to Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.

Indicator 39 - Public debt evolution — domestic/externall stock, currency, maturity and instrument - Overview

Source	MINFIN; UGD
Format	PDF
Periodicity	Quarterly
Module/Tier	M1.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_2_public_debt_evolution_domestic_externall
Sheet/section/columns to read	Extract the tables 1 and 6 from the PDF | Pay attention to fields in bold because they may be calculated fields; these do not need to be loaded
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the public debt statistical bulletin, tables 1 and 6.

Objective: Make available stock, disbursements, redemptions and main government debt ratios.

Persona: Public debt / sovereign risk analyst

2 Development scope

In scope

- Configure the indicator 39 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Extract the tables 1 and 6 from the PDF | Pay attention to fields in bold because they may be calculated fields; these do not need to be loaded
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Public debt evolution — domestic/externall stock, currency, maturity and instrument - Overview into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 39, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.

- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 39 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='39' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Extract the tables 1 and 6 from the PDF | Pay attention to fields in bold because they may be calculated fields; these do not need to be loaded
- Expected periodicity: Quarterly; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- Extract Table 1 and Table 6 from the PDF. Indicators in bold or calculated fields are not included in the main observations; derived ratios are handled in Silver/Gold if necessary. In Bronze, persist only raw values and metadata; final calculations belong to Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.
- PDF without table extractable or with low confidence: record PDF_EXTRACTION_LOW_CONFIDENCE and encaminhar for validation manual.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Cross-Series Consistency: run informative validation of accounting/fiscal identities when published totals exist; flag only.

Indicator 40 - Main Activity Indicators

Source	Commercial Banks
Format	XLSX/PDF
Periodicity	Quarterly / Annual
Module/Tier	M2.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m2_2_main_activity_indicators
Sheet/section/columns to read	Load historys and values atuais of the Annual Reports and Accounts. Normalmente the table aparece after the Mensagem of the Conselho of Administration.
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the main indicators of activity of the commercial banks based on the respective Annual Reports and Accounts.

Objective: Allow comparison of scale, operational evolution and commercial dynamics by banking institution.

Persona: Banking / financial activity analyst

2 Development scope

In scope

- Configure the indicator 40 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load historys and values atuais of the Annual Reports and Accounts. Normalmente the table aparece after the Mensagem of the Conselho of Administration.
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Main Activity Indicators into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 40, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 40 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='40' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load historys and values atuais of the Annual Reports and Accounts. Normalmente the table aparece after the Mensagem of the Conselho of Administration.
- Expected periodicity: Quarterly / Annual; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify source as Tier 2 — Institutional sector source / official publication of the bank and keep this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.
- Subcomponents/line items functional to observe: assets; credit; deposits; customers; operations; other KPIs of activity when published.

Formulas

- Bronze must persist the values published by bank and period. Ratios, rankings and comparative sector will be derived in Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP, connector, URL or collection folder: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.
- PDF without table extractable or with low confidence: record PDF_EXTRACTION_LOW_CONFIDENCE and encaminhar for validation manual.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 41 - Balance Sheet and Income Statement by Bank (summary)

Source	BNA; Commercial banks
Format	XLSX/PDF
Periodicity	Quarterly / Annual
Module/Tier	M2.3 / Detail
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m2_3_balance_sheet_and_income_statement_by_bank_summary
Sheet/section/columns to read	Load historys and values atuais of the Annual Reports and Accounts. Normalmente the information aparece before of the Notas to the Financial statements Financial.
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the balance sheet and of the income statement by bank, in view resumida.

Objective: Create comparable base for assets, liabilities, capital, revenues, costs and net profit/loss by institution.

Persona: Banking financial statements / risk and performance analyst

2 Development scope

In scope

- Configure the indicator 41 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load historys and values atuais of the Annual Reports and Accounts. Normalmente the information aparece before of the Notas to the Financial statements Financial.
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Balance Sheet and Income Statement by Bank (summary) into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 41, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 41 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='41' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load historys and values atuais of the Annual Reports and Accounts. Normalmente the information aparece before of the Notas to the Financial statements Financial.
- Expected periodicity: Quarterly / Annual; update SLA: up to 10 business days after official publication; validate quarterly sequence.
- Expected period fields: reference_year, reference_quarter, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.
- Subcomponents/line items functional to observe: balance sheet; income statement; bank; period; line item accounting.

Formulas

- Bronze must preserve accounting line items as published. Financial ratios, margins and rankings remain outside Bronze.

Error scenarios and expected handling

- File not found in the ZIP, connector, URL or collection folder: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.
- PDF without table extractable or with low confidence: record PDF_EXTRACTION_LOW_CONFIDENCE and encaminhar for validation manual.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Range/Outlier Validation: mark values outside plausible historical limits the warning or critical according to the indicator.

Indicator 42 - Domestic sovereign yield curve

Source	UGD
Format	PDF
Periodicity	Weekly / Monthly
Module/Tier	M3.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m3_2_domestic_sovereign_yield_curve
Sheet/section/columns to read	Load the Daily Bulletin table related to the Treasury Securities Yield Curve, preserving date, maturity, yield and instrument.
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the domestic sovereign yield curve by maturity.

Objective: Make available structure temporal of yields of the securities of the Treasury for analysis of market of debt and cost of financing.

Persona: Public debt / equity market / sovereign risk analyst

2 Development scope

In scope

- Configure the indicator 42 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load the Daily Bulletin table related to the Treasury Securities Yield Curve, preserving date, maturity, yield and instrument.
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Domestic sovereign yield curve into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 42, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 42 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='42' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load the Daily Bulletin table related to the Treasury Securities Yield Curve, preserving date, maturity, yield and instrument.
- Expected periodicity: Weekly / Monthly; update SLA: Up to 5 business days after official publication; validate weekly sequence when applicable.
- Expected period fields: reference_year, reference_week, reference_date.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.
- Subcomponents/line items functional to observe: maturity; yield; security; data of the curve.

Formulas

- Bronze must store as published curve points. Interpolation, curve shifts and spread calculations will be performed in Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP, connector, URL or collection folder: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.
- PDF without table extractable or with low confidence: record PDF_EXTRACTION_LOW_CONFIDENCE and encaminhar for validation manual.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Cross-Series Consistency: run informative validation between series related when totals, ratios or components are published; flag only.

Indicator 43 - Corporate bonds and commercial paper/ IPO

Source	BODIVA; CMC
Format	PDF/WEB
Periodicity	whenever updated
Module/Tier	M3.3 / Detail
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m3_3_corporate_bonds_and_commercial_paper_ipo
Sheet/section/columns to read	Load data when updated in the BODIVA/CMC sources, identifying instrument, issuer, data, amount, price/rate and state of the operation.
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of corporate issuances, papel commercial and ofertas public/IPO published by BODIVA or CMC.

Objective: Monitor the development of the private equity market and new corporate financing instruments.

Persona: Equity market / corporate finance analyst

2 Development scope

In scope

- Configure the indicator 43 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load data when updated in the BODIVA/CMC sources, identifying instrument, issuer, data, amount, price/rate and state of the operation.
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Corporate bonds and commercial paper/ IPO into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 43, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.
- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 43 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='43' and indicator_candidate_name populated.

- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load data when updated in the BODIVA/CMC sources, identifying instrument, issuer, data, amount, price/rate and state of the operation.
- Expected periodicity: whenever updated; SLA of update: According to periodicity published by the source; record publication date and extraction date.
- Expected period fields: reference_period, reference_date when available.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.
- Subcomponents/line items functional to observe: bonds corporate; papel commercial; IPO; issuer; amount; data; status.

Formulas

- Bronze must store published events and instruments. Aggregations by issuer, amount total and series historical will be derived in Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP, connector, URL or collection folder: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.
- PDF without table extractable or with low confidence: record PDF_EXTRACTION_LOW_CONFIDENCE and encaminhar for validation manual.
- Web page/API unavailable or blocked: record SOURCE_UNAVAILABLE, save a snapshot when possible and apply controlled retry.
- Source without update in the period: record NO_NEW_PUBLICATION without failurer the execution.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Cross-Series Consistency: run informative validation between series related when totals, ratios or components are published; flag only.

Indicator 44 - Economic Diversification Index

Source	Laza
Format	Indicator calculated
Periodicity	Semiannual
Module/Tier	M4.2 / Extended
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m4_2_economic_diversification_index
Sheet/section/columns to read	Formalize calculation methodology, source base, weights, formula, methodological version and periodicity. Load approved sector inputs and index result.
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the inputs and profit/loss of the Economic Diversification Index, indicator calculated internally by the LAZA.

Objective: Provide the synthetic metric of economic diversification, with traceable methodology and versioned.

Persona: Product Owner / analyst strategy / team of economy applied

2 Development scope

In scope

- Configure the indicator 44 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Formalize calculation methodology, source base, weights, formula, methodological version and periodicity. Load approved sector inputs and index result.
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Economic Diversification Index into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 44, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.

- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 44 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='44' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Formalize calculation methodology, source base, weights, formula, methodological version and periodicity. Load approved sector inputs and index result.
- Expected periodicity: Semiannual; update SLA: Up to 15 business days after fechamento semiannual or approval methodological.
- Expected period fields: reference_year, reference_semester, reference_date.
- Classify the source as an internal LAZA calculated indicator — requires versioned methodology — and keep this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.
- Subcomponents/line items functional to observe: index calculated; sector inputs; weights; methodological version; profit/loss semiannual.

Formulas

- Indicator calculated: the Bronze must store inputs, parameters and outputs approved. The formula and weights must be versioned; recalculation and official publication of the index occur in Silver/Gold according to contract methodological.

Error scenarios and expected handling

- File not found in the ZIP, connector, URL or collection folder: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- Cross-Series Consistency: run informative validation between series related when totals, ratios or components are published; flag only.

Indicator 45 - Balance Sheet Center by sector of activity

Source	BNA
Format	XLSX
Periodicity	Annual
Module/Tier	M1.3 / Detail
Physical Bronze target	bronze.raw_indicator_observation
Logical subject / optional view	brz_m1_3_balance_sheet_center_by_sector_of_activi
Sheet/section/columns to read	Load columns C to I - exclude totals and fields that may be calculated; Load columns C to H - exclude totals and fields that may be calculated
Bronze granularity	One row in raw_indicator_observation per eligible raw extracted value from the spreadsheet, PDF, API or source table, preserving physical location and metadata.

1 Overview and context

Summary: Ingestion of the Centers of Balance Sheet by sector of activity.

Objective: Make available indicators sector and pyearramas for analysis corporate and macro sector.

Persona: Sector analyst / balance-sheet-by-activity analyst

2 Development scope

In scope

- Configure the indicator 45 in the ingestion table/configuration with name, source, format, periodicity, logical subject, reading rule and SLA.
- Register batch, file, version and hash before data extraction.
- Read the defined source for the indicator, especially: Load columns C to I - exclude totals and fields that may be calculated; Load columns C to H - exclude totals and fields that may be calculated
- Populate bronze.raw_indicator_observation with the eligible raw values, keeping indicator_candidate_code, indicator_candidate_name, source_file_id, file_version_id, batch_id, sheet_name/pdf_page_number, row_number, column_name/cell_reference, raw_period, raw_unit, raw_value_text, raw_value_numeric when possible, raw_record_json and raw_record_hash.
- Register technical events in raw_parse_event and invalid records in raw_quarantine_record.
- Apply Bronze quality checks: source provenance, completeness, duplication detection, numeric parse, unit/scale, temporal consistency and timeliness.

Out of scope

- Calculate variations, final aggregations, scores or derived metrics that are not explicitly published by the source.
- Resolve conflicts between sources, harmonize dimensions or create definitive Data Vault business keys.
- Publish dashboards, executive cards or the dimensional consumption model.
- Mannually correct values without recording the event, reason and governance trail.

3 Functional requirements and user story

User Story: As a LAZA I-Hub Data Analyst/Data Engineer, I want to load the indicator Balance Sheet Center by sector of activity into the Bronze layer with provenance, batch, hash, physical location and technical validations so that the Silver and Gold layers can consume traceable and auditable raw data.

Acceptance criteria

- Given the file/source for indicator 45, when the pipeline runs, it must create or update raw_ingestion_batch with batch_id, status, start/end timestamp, source and execution metrics.
- Given that the file is available, when the load starts, raw_source_file and raw_file_version must contain file_hash, path/URL, format, size, institution, publication and extraction date.

- Given that the expected sheet/section/columns exist, when extraction finishes, all eligible observations for indicator 45 must be stored in bronze.raw_indicator_observation with indicator_candidate_code='45' and indicator_candidate_name populated.
- Each observation must have source_file_id, file_version_id, batch_id, raw_value_text, raw_record_json, raw_record_hash, load_dts and record_source.
- When available, the observation must keep sheet_name or pdf_page_number, row_number, column_name and cell_reference for physical traceability.
- The load must be idempotent: reprocessing the same file with the same hash cannot create an active duplication.
- If a file is missing, a sheet/section does not exist, the layout changes, an unexpected non-numeric value appears or the period is invalid, the event must be recorded in parse_event/quarantine with severity and reason.
- The execution must calculate parse_status and dq_flags_json without preventing storage of the raw file.

4 Business rules, formulas and edge cases

Business rules

- Validation spreadsheet rule: Load columns C to I - exclude totals and fields that may be calculated
- Validation spreadsheet rule: Load columns C to H - exclude totals and fields that may be calculated
- Expected periodicity: Annual; update SLA: Up to 15 business days after official publication; validate the reference year.
- Expected period fields: reference_year.
- Classify the source as Tier 1 - primary official source and maintain this classification in each observation.
- Preserve negative values, signs, decimal places, thousand separators and the unit/scale reported by the source.

Formulas

- For each sector file, load Sector Indicators columns C:I and Sector Program columns C:H. Exclude totals and calculated fields. In Bronze, persist only raw values and metadata; final calculations belong to Silver/Gold.

Error scenarios and expected handling

- File not found in the ZIP or connector: log FILE_NOT_FOUND, keep the batch as failed and notify the Product Owner/Data Engineer.
- Sheet/section/table not found: register SHEET_OR_SECTION_NOT_FOUND and attach the list of available sheets/sections to the error event.
- Layout change: register LAYOUT_CHANGED when expected headers, numeric columns or period patterns are not recognized.
- Invalid or non-numeric value in the field expected to be numeric: write raw_value_text, leave raw_value_numeric null and create dq_flag NUMERIC_PARSE_ERROR.
- Missing or duplicate period: register PERIOD_MISSING or DUPLICATE_PERIOD and prevent automatic promotion to Silver.
- Data delayed against periodicity: register FRESHNESS_WARNING or STALE_DATA according to the tolerance defined by the DQF.

Applicable quality checks

- Completeness: verify whether all expected periods are present according to the periodicity.
- Duplicate Detection: detect duplicates by indicator_id + period + source + file + item/dimension.
- Unit/Scale Validation: validate that the unit/scale has not changed between batches without documentation.
- Temporal Consistency: flag extreme variations for review, without automatically correcting them in Bronze.
- Source Provenance: verify that source, file, hash and extraction method have been populated.
- No critical event open for the most recent batch.
- Completeness compatible with periodicity expected or exception approved by the Product Owner.
- Metadata of provenance fulls for 100% of the observations active.
- Hash of file registrado and reprocessing idempotente validated.
- Rules of excluare of totals/subtotals/calculated applied according to Validation.

- Documentation methodological created or currentizada for the Data Sheet of the indicator.

TESTES E DEFINITION OF DONE

9. Tests functional minimum

Teste	Input	Profit/loss expected
Load valid XLSX	QUARTERLY NATIONAL ACCOUNTS.xlsx	Batch SUCCEEDED, registered sheets and generated raw observations.
Load valid PDF	Indicators of Solidez Financeira.pdf	registered pages, extracted tables or documented PARTIAL events.
Duplicidade	Mesmo file both timees	Segundo processamento not duplica observations active.
Schema changed	Sheet with columns different	Evento WARNING/CRITICAL and quarantine when necessary.
Reprocessing	Batch reexecutado	Records manhave idempotency and lineage Consolidated.

9.1 Tests additional by indicator

- Run by the menos the load positiva by format from the source: XLSX/XLS, PDF and API when applicable.
- Validate that all indicators included have records of configuration and logical subject.
- Validate that indicators 27, 32, 35, 37, 40, 41, 42, 43 and 44 are registered, processable and generate active observations when a source is available.
- Validate that file layout changes generate parse_event and do not break the entire pipeline.
- Validate that raw_record_hash is stable across reprocessing of the same content.
- Validate that quarantined data remains linked to batch_id, source_file_id and file_version_id.

10. Checklist of Definition of Done

- Bronze tables created in Delta and versioned in the code repository.
- Main table bronze.raw_indicator_observation created with all mandatory fields.
- Tables raw_ingestion_batch, raw_source_file, raw_file_version, raw_excel_sheet, raw_pdf_page, raw_extracted_row/cell, raw_parse_event and raw_quarantine_record available or mapeadas for equivalent approved.
- Configuration of all the 45 eligible indicators cadastrada and versioned.
- Pipelines Excel, PDF and API with batch logging, hash, quarantine and reprocessing controlled.
- Raw observations traceable to the physical source.
- Metrics of load and technical events persistidos.
- Unit and integration tests covering duplication, parsing error, missing sheet, invalid value, layout change and reprocessing.
- Operational documentation available: execution parameters, reprocessing instructions, lineage query and quarantine handling.
- Functional approval by Data Analyst/Data Steward before promotion to Silver Raw Vault.

APPENDIX

11. Backlog technical suggested for implementation

Topic	Story / task	Deliverable expected
BRZ-EPIC-01 Configuration	Create table/configuration from the sources and Validation indicators.	Config validated with 36 indicators assets and list of excluded.
BRZ-EPIC-02 Source Registry	Implement record of file, version, hash and path/URL.	raw_source_file and raw_file_version populadas by load.
BRZ-EPIC-03 Parsers	Implement parser Excel and PDF with extraction of sheet/page/row/cell.	raw_excel_sheet/raw_pdf_page/raw_extracted_* preenchidas.
BRZ-EPIC-04 Observations	Transformsr extractions in raw_indicator_observation.	Observations by indicator with raw_value_text

		and lineage physical.
BRZ-EPIC-05 DQ technical	Implement checks Bronze and flags of quality.	parse_status, dq_flags_json and events persistidos.
BRZ-EPIC-06 Quarantine	Persist records invalid without block file raw.	raw_quarantine_record with severity and reason.
BRZ-EPIC-07 Reprocessing	Implement idempotency by file_hash and raw_record_hash.	Reprocessing without duplication active.
BRZ-EPIC-08 Observabilidade	Publish metrics of load, contagens and status of batch.	Viare operational for suporte and audit.